

Scenario A: Correct Client Certificate

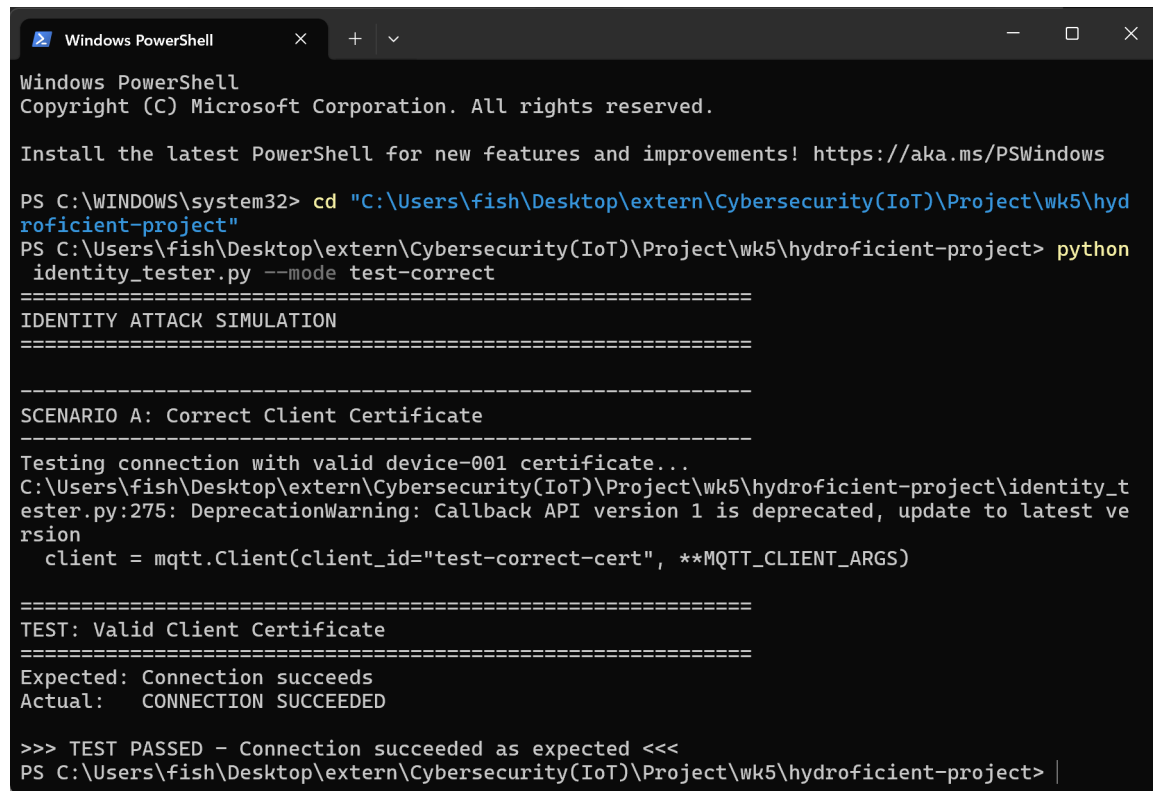
Goal: Verify that normal mTLS connections work.

Field	Your Result
Connection outcome	Succeeded
Error message (if any)	n/a
Expected outcome	Connection succeeds
Test result	PASS

Analysis: Did the connection succeed as expected?

Yes, the mTLS connection succeeds.

Screenshot: (Paste your screenshot below)



```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> cd "C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project"
PS C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project> python identity_tester.py --mode test-correct
=====
IDENTITY ATTACK SIMULATION
=====

-----
SCENARIO A: Correct Client Certificate
-----

Testing connection with valid device-001 certificate...
C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project\identity_tester.py:275: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
    client = mqtt.Client(client_id="test-correct-cert", **MQTT_CLIENT_ARGS)

=====
TEST: Valid Client Certificate
=====
Expected: Connection succeeds
Actual: CONNECTION SUCCEEDED

>>> TEST PASSED - Connection succeeded as expected <<<
PS C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project> |
```

Scenario B: No Client Certificate

Goal: Verify that the broker rejects devices without certificates.

Field	Your Result
Connection outcome	Failed
Error message (if any)	Peer did not return a certificate
Expected outcome	Connection rejected
Test result	PASS

Analysis: What specific error did you see? This error proves that a rogue device cannot connect just by having network access.

I got the error message "peer didn't return a certificate". It means the broker asked for a client cert, but didn't get one. So that means rogue devices cannot connect our system even though they have network access.

Screenshot: (Paste your screenshot below)

The image contains two side-by-side screenshots of Windows PowerShell windows. The left window shows the execution of a Python script named 'identity_tester.py' in the directory 'C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project'. The script output indicates 'SCENARIO B: No Client Certificate' and 'Testing connection WITHOUT any client certificate...'. It shows a deprecation warning for the 'mqtt.Client' function and then reports 'TEST: No Client Certificate', 'Expected: Connection rejected', 'Actual: CONNECTION FAILED', and 'Error: Connection failed with rc=-1'. The final output is '>>> TEST PASSED - Connection was correctly rejected <<<'. The right window shows the execution of 'mosquitto -c mosquitto_mtls.conf -v' in the same directory. The output shows 'mosquitto version 2.1.2 starting', 'Config loaded from mosquitto_mtls.conf', 'Bridge support available', 'Persistence support available', 'TLS support available', 'TLS-PSK support available', 'Websockets support available', 'Opening ipv6 listen socket on port 8883', 'Opening ipv4 listen socket on port 8883', and 'mosquitto version 2.1.2 running'. It then shows two new connections from '::1:60977' and '::1:60980' on port 8883, both resulting in 'error:0A0000C7:SSL routines:peer did not return a certificate' and 'Client ::1 [::1:60977] disconnected: protocol error'.

```
PS C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project> python identity_tester.py --mode test-no-cert

IDENTITY ATTACK SIMULATION
=====

SCENARIO B: No Client Certificate

Testing connection WITHOUT any client certificate...
C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project\identity_tester.py:314: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client(client_id="test-no-cert", **MQTT_CLIENT_ARGS)

=====
TEST: No Client Certificate
=====
Expected: Connection rejected
Actual:   CONNECTION FAILED
Error:    Connection failed with rc=-1

>>> TEST PASSED - Connection was correctly rejected <<<
PS C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project>
```

```
PS C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project> mosquitto -c mosquitto_mtls.conf -v
1785446814: mosquitto version 2.1.2 starting
1785446814: Config loaded from mosquitto_mtls.conf.
1785446814: Bridge support available.
1785446814: Persistence support available.
1785446814: TLS support available.
1785446814: TLS-PSK support available.
1785446814: Websockets support available.
1785446814: Opening ipv6 listen socket on port 8883.
1785446814: Opening ipv4 listen socket on port 8883.
1785446814: mosquitto version 2.1.2 running
1785446827: New connection from ::1:60977 on port 8883.
1785446827: OpenSSL Error while trying to get the error[0]: error:0A0000C7:SSL routines:peer did not return a certificate
1785446827: Client ::1 [::1:60977] disconnected: protocol error.
1785446828: New connection from ::1:60980 on port 8883.
1785446828: OpenSSL Error while trying to get the error[0]: error:0A0000C7:SSL routines:peer did not return a certificate
1785446828: Client ::1 [::1:60980] disconnected: protocol error.
```

Scenario C: Certificate from Wrong CA

Goal: Verify that only certificates signed by your CA are accepted.

Field	Your Result
Connection outcome	Failed
Error message (if any)	SSL routines: certificate verify failed
Expected outcome	Connection rejected
Test result	PASS

Analysis: How does this differ from Scenario B? Both are rejected, but for different reasons.

I got the error message "SSL routines: certificate verification failed". This error reveals it has a certificate to request connection, but it is not coming from the trust one. Compared with scenario B, this error reveals it has a client certificate, but not the trust one. In scenario B, that one is no client certification to verify.

Screenshot: (Paste your screenshot below)

The screenshot displays two overlapping Windows PowerShell windows. The left window shows the execution of a Python script named `identity_tester.py` with the `--mode test-wrong-ca` argument. The script output includes a header "IDENTITY ATTACK SIMULATION", a section for "SCENARIO C: Certificate from Wrong CA", and a test result "TEST: Wrong CA Certificate" which shows the connection was correctly rejected. The right window shows the output of the `mosquitto -c mosquitto_mtls.conf -v` command, displaying the Mosquitto version 2.1.2 starting, configuration loaded, and various support features available. It also shows two client connections being rejected with the error "SSL routines: certificate verify failed".

```
PS C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project> python identity_tester.py --mode test-wrong-ca

IDENTITY ATTACK SIMULATION

=====

SCENARIO C: Certificate from Wrong CA

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Generating rogue CA and client certificate...
Done. Rogue CA and device certificate created.
Testing connection with certificate signed by DIFFERENT CA..

C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project\identity_tester.py:359: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client(client_id="test-wrong-ca", **MQTT_CLIENT_ARGS)

=====

TEST: Wrong CA Certificate
=====
Expected: Connection rejected
Actual:   CONNECTION FAILED
Error:    Connection failed with rc=-1

>>> TEST PASSED - Connection was correctly rejected <<<
PS C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project>
```

```
PS C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project> mosquitto -c mosquitto_mtls.conf -v
1785446974: mosquitto version 2.1.2 starting
1785446974: Config loaded from mosquitto_mtls.conf.
1785446974: Bridge support available.
1785446974: Persistence support available.
1785446974: TLS support available.
1785446974: TLS-PSK support available.
1785446974: Websockets support available.
1785446974: Opening ipv6 listen socket on port 8883.
1785446974: Opening ipv4 listen socket on port 8883.
1785446974: mosquitto version 2.1.2 running
1785446999: New connection from ::1:64593 on port 8883.
1785446999: OpenSSL Error while trying to get the error[0]: error:0A000086:SSL routines::certificate verify failed
1785446999: Client ::1 [::1:64593] disconnected: protocol error.
1785447000: New connection from ::1:64596 on port 8883.
1785447000: OpenSSL Error while trying to get the error[0]: error:0A000086:SSL routines::certificate verify failed
1785447000: Client ::1 [::1:64596] disconnected: protocol error.
```

Scenario D: Expired Certificate

Goal: Verify that expired certificates are rejected.

Field	Your Result
Connection outcome	Failed
Error message (if any)	SSL routines: certificate verify failed
Expected outcome	Connection rejected
Test result	PASS

Analysis: Why is certificate expiration important for IoT security? What happens if a device certificate never expires?

I got the error message "SSL routines: certificate verification failed". It reveals the existing certificate was rejected due to an expired certificate or wrong certificate. It is important to set the expiration date of the certificate. If the device certificate never expires, someone didn't work in the same company. They still can use the same certificate to access the system. In addition, new certificates of certification happen every year, setting up expired certificates is another way to keep the updated version of certificate for protecting the system.

Screenshot: (Paste your screenshot below)

```
Record your results in the Scenario C table in your copied template.

Windows PowerShell
=====
SCENARIO D: Expired Certificate
=====
Generating expired client certificate...
Done. Expired certificate created (expired yesterday).
Testing connection with EXPIRED client certificate...
C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project\identity_tester.py:412: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client(client_id="test-expired", **MQTT_CLIENT_ARGS)
=====
TEST: Expired Certificate
=====
Expected: Connection rejected
Actual: CONNECTION FAILED
Error: Connection failed with rc=-1

>>> TEST PASSED - Connection was correctly rejected <<<
PS C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project>

Windows PowerShell
PS C:\Users\fish\Desktop\extern\Cybersecurity(IoT)\Project\wk5\hydroficient-project> mosquitto -c mosquitto_mtls.conf -v

1785494028: mosquitto version 2.1.2 starting
1785494028: Config loaded from mosquitto_mtls.conf.
1785494028: Bridge support available.
1785494028: Persistence support available.
1785494028: TLS support available.
1785494028: TLS-PSK support available.
1785494028: Websockets support available.
1785494028: Opening ipv6 listen socket on port 8883.
1785494028: Opening ipv4 listen socket on port 8883.
1785494028: mosquitto version 2.1.2 running
1785494038: New connection from ::1:57993 on port 8883.
1785494038: OpenSSL Error while trying to get the error[0]: error:0A000086:SSL routines::certificate verify failed
1785494038: Client ::1 [::1:57993] disconnected: protocol error.
1785494039: New connection from ::1:57996 on port 8883.
1785494039: OpenSSL Error while trying to get the error[0]: error:0A000086:SSL routines::certificate verify failed
1785494039: Client ::1 [::1:57996] disconnected: protocol error.
```

Summary Worksheet

Complete this summary of your identity attack simulations:

Scenario	What You Tried	Expected	Actual	Pass/Fail
A: Correct cert	Connect with valid device-001 cert	Success	Success	Pass
B: No cert	Connect without client certificate	Rejected	Rejected	Pass
C: Wrong CA	Connect with cert from different CA	Rejected	Rejected	Pass
D: Expired	Connect with expired certificate	Rejected	Rejected	Pass